INCH-POUND
MIL-DTL-24643/49G
27 November 2023
SUPERSEDING
MIL-DTL-24643/49F(SH)
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/49F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSTNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation (see table I for thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, the insulation color shall be black.
Fourth	-	Three conductors cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:14Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/49	Type and size	Conductor size		Insulation thickness min. avg. (inch)	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
		Navy STD	AWG/ MCM/			min.	max.
-01UN	LSTNW-3	--	16 (class B)	0.030	0.040	0.380	0.411
-02UN	LSTNW-4	--	14 (class B)	0.030	0.040	0.415	0.449
-03UN	LSTNW-9	--	10 (class B)	0.030	0.050	0.578	0.625
-04UN	LSTNW-14	--	9 (class B)	0.045	0.050	0.620	0.670
-05UN	LSTNW-23	--	7 (class B)	0.045	0.055	0.703	0.760
-055UN	LSTNW-30	--	5 (class C)	0.045	0.055	0.799	0.865
-06UN	LSTNW-50	--	3 (class C)	0.045	0.060	0.896	0.969
-07UN	LSTNW-75	--	1 (class C)	0.055	0.070	1.048	1.134
-08UN	LSTNW-100	--	0 (class D)	0.055	0.075	1.171	1.266
-09UN	LSTNW-150	--	000 (class D)	0.055	0.075	1.401	1.515
-10UN	LSTNW-200	--	0000 (class D)	0.055	0.075	1.544	1.669
-11UN	LSTNW-300	--	300 MCM (class D)	0.065	0.090	1.810	1.957
-12UN	LSTNW-400	400 (127)	--	0.065	0.090	2.038	2.203

TABLE II. Electrical details.

Type and size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand min. (volts)	Accelerated service loading current (amperes)
LSTNW-3	4.3	6,000	4,000	N/A
LSTNW-4	2.68	5,000	4,000	N/A
LSTNW-9	1.08	3,800	4,000	N/A
LSTNW-14	0.859	4,400	5,500	90
LSTNW-23	0.543	3,700	5,500	115
LSTNW-30	0.358	3,000	5,500	140
LSTNW-50	0.21	2,600	5,500	185
LSTNW-75	0.134	2,600	7,000	250
LSTNW-100	0.106	2,400	7,000	295
LSTNW-150	0.0674	1,800	7,000	400
LSTNW-200	0.053	1,500	7,000	460
LSTNW-300	0.0377	1,200	7,000	590
LSTNW-400	0.0273	1,000	7,000	750

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:14Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table II
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	See table II
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table II
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:14Z
Check the source to verify that this is the current version before use.

Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation), method 1 only (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Accelerated service, sizes 14 and larger (see table II for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:14Z
Check the source to verify that this is the current version before use.

Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSTNW	3	2,500
LSTNW	4, 9, and 14	2,000
LSTNW	23 and 30	1,500
LSTNW	50, 75, 100, and 150	1,000
LSTNW	200, 300, and 400	800

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-065)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:14Z
Check the source to verify that this is the current version before use.